

YASH PAREKH

helloyash789@gmail.com | +44 7386 488821 | London, UK
github.com/YashParekh14 | linkedin.com/in/yash-parekh1427

SUMMARY

Software engineer with production experience in full stack development (MERN) and Artificial Intelligence / Machine Learning engineering. Currently completing MSc Artificial Intelligence (full-time) at De Montfort University, Leicester (Jan 2025 - Jun 2026). Built and deployed three AI systems: a Retrieval-Augmented Generation (RAG) pipeline with evaluation harness, a production agentic shopping assistant with CI/CD on AWS, and a medical Vision Transformer achieving 95.06% accuracy. Focused on Large Language Model (LLM) application development, retrieval systems, and end-to-end AI deployment.

PROJECTS

Telecom RAG Chatbot | Python | LangChain | ChromaDB | GPT-4o-mini | Streamlit | Docker [\[GitHub\]](#)

- Built a conversational customer support chatbot using Retrieval-Augmented Generation (RAG) retrieving context from three sources: FAQ CSV, resolved support tickets (SQLite), and PDF guide; implemented cross-encoder reranking (ms-marco-MiniLM), score-threshold filtering, and deduplication instead of blind top-k retrieval
- Developed LLM-as-judge evaluation harness measuring retrieval hit rate and answer faithfulness; identified 2 knowledge gaps through evaluation output, augmented FAQ data, improved retrieval hit rate from 83% to 100%; average faithfulness score 4.83/5 — experiments confirmed data quality beats hyperparameter tuning
- Implemented conversation memory with query condensation for coherent multi-turn dialogue; 10 pytest unit tests; Dockerised and deployed live on Streamlit Community Cloud

AI Shopping Assistant - Agentic Tool Use | Python | GPT-4o | LangChain | PostgreSQL | Docker | AWS EC2 | GitHub Actions | LangSmith [\[GitHub\]](#)

- Built conversational shopping agent using manual ReAct tool-call loop replacing LangChain built-in agent which silently dropped tool results; eliminated N+1 query pattern by joining product ratings at SQL level, reducing multiple LLM round-trips to a single tool call
- Dual-backend database architecture: SQLite for development, PostgreSQL on Supabase for production, switchable via single environment variable; GPT-4o vision API for product image identification; checkout validation in application code prevents hallucinated product IDs from creating invalid orders
- Full CI/CD pipeline via GitHub Actions: runs 15 unit tests and 15 behavioural evaluation scenarios on every push, builds Docker image with commit SHA tag, deploys to AWS EC2 (eu-west-2 London); LangSmith observability tracking every tool call, token count, latency, and cost (~\$0.01 per conversation)

NeuroScan AI - Brain Tumor Detection | PyTorch | Vision Transformer | FastAPI | React | Docker | MLflow | Hugging Face [\[GitHub\]](#)

- Systematic model comparison: EfficientNet-B0 baseline 84.38% accuracy; frozen Vision Transformer (ViT) scored 75.25%; unfreezing last 4 of 12 transformer blocks with AdamW optimiser (lr=5e-6) and CosineAnnealingLR scheduler achieved 95.06% accuracy and 0.9866 AUC-ROC on 7,023 brain MRI images — +19.81% improvement over frozen ViT
- Applied MixUp, CutMix, and RandomErasing augmentation pipeline with label smoothing; Grad-CAM explainability confirms model attention on tumor regions; 30 experiments tracked with MLflow; FastAPI backend on Hugging Face Spaces (Docker) + React frontend on Vercel with GitHub CI/CD auto-deployment

EXPERIENCE

AI/ML Intern | NeuronPath - Virtual Internship

Apr 2026 - May 2026

- Completed structured Artificial Intelligence and Machine Learning internship covering supervised learning, model evaluation, and applied Natural Language Processing (NLP); implemented Python projects using scikit-learn and TensorFlow on real-world datasets

Warehouse Operative | Vittles Food Ltd, UK (Part-time)

Oct 2025 - Mar 2026

- Part-time role undertaken alongside full-time MSc Artificial Intelligence studies; order picking, stock management, and quality control in bakery warehouse environment

MERN Stack Developer | eVitalRx

Mar 2023 - Jan 2025

- Built RESTful APIs and React user interfaces for pharmacy management platform; owned backend features end-to-end from database schema design to production deployment using MongoDB, Express, React, and Node.js stack

- Collaborated in hybrid production team with code review and testing practices; platform served pharmacies across India

CERTIFICATIONS

- Machine Learning Fundamentals - Cohort Alpha (Feb 2026)
- Data Science Essentials - Cohort Alpha (Apr 2026)
- Generative AI Foundations and Applications - Cohort Alpha (Jun 2026)

TECHNICAL SKILLS

AI / LLM: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), ChromaDB, Vector Search, Cross-encoder Reranking, Agentic AI, Prompt Engineering, LLM Evaluation, LangSmith

Machine Learning / Deep Learning: PyTorch, TensorFlow, Vision Transformer (ViT), Convolutional Neural Networks (CNN), EfficientNet, Scikit-learn, Natural Language Processing (NLP), NumPy, Pandas, Matplotlib, MLflow

Languages and Stack: Python, JavaScript, SQL, React, Node.js, Express, MongoDB, FastAPI, PostgreSQL

Tools and Cloud: Docker, GitHub Actions, AWS EC2, Streamlit, Hugging Face, Vercel, Git, Postman

EDUCATION

MSc Artificial Intelligence (Full-time) De Montfort University, Leicester, UK	Jan 2025 - Jun 2026
BE Information Technology Gandhinagar Institute of Technology, India	2017 - 2021